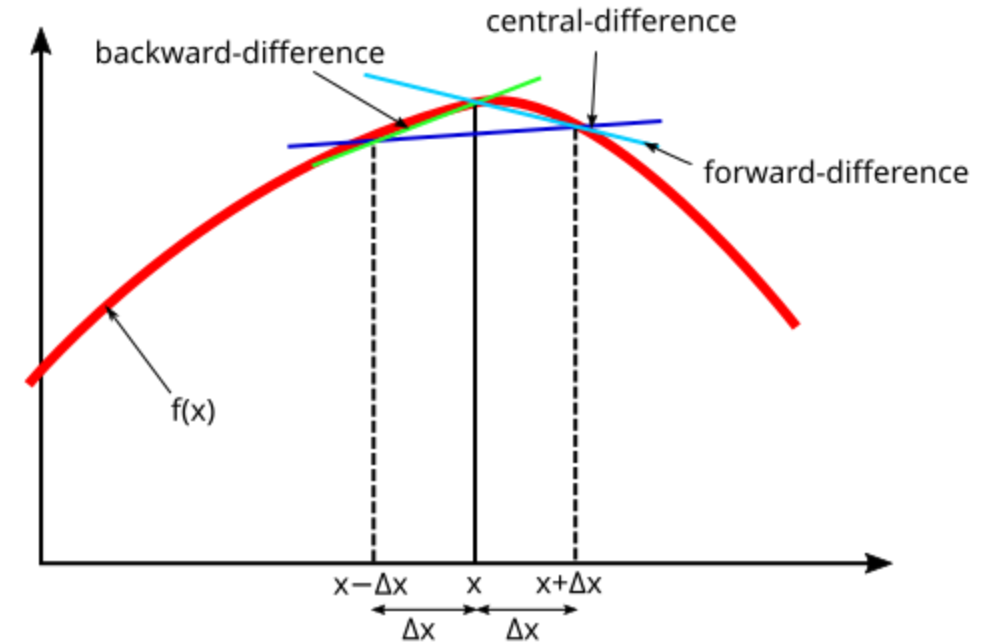


Day 03 - Getting Started with VS Code

$$v_i \approx \frac{x_{i+1} - x_i}{\Delta t}$$

Today we'll set up VS Code for scientific computing and practice numerical differentiation.

PHY 321 Classical Mechanics I - Spring 2026



Today's Goals

After this session, you should be able to:

- **Set up VS Code** for Python scientific computing
- **Use Jupyter notebooks** in VS Code
- **Perform numerical differentiation** using finite difference methods
- **Compare** numerical results with analytical solutions

Step 1: Download VS Code

Go to: <https://code.visualstudio.com/download>

Choose your operating system:

- **macOS**: Universal download (Intel + Apple Silicon)
- **Windows**: User Installer
- **Linux**: .deb, .rpm, or .tar.gz

The download is ~200 MB and takes just a minute or two.

Step 2: Install VS Code

On macOS:

1. Open the downloaded `.zip` file
2. Drag `Visual Studio Code.app` to **Applications**
3. Launch from Applications

On Windows:

1. Double-click the `.exe` installer
2. Follow the wizard (default settings are fine)
3. VS Code launches automatically

Step 3: Install Required Extensions

Click the **Extensions** icon (4 squares) on the left sidebar

- Or press `Ctrl+Shift+X` (Windows/Linux) or `Cmd+Shift+X` (Mac)

Install these two extensions:

1. **Jupyter** by Microsoft
2. **Python** by Microsoft

Click **Reload** when prompted.

Step 4: Verify Python

Open VS Code's Terminal: **View** → **Terminal** (or `Ctrl+``)

Type:

```
python --version
```

or

```
python3 --version
```

You should see something like `Python 3.10.0` or newer.

If not, install Python from: <https://www.python.org/downloads/>

Download Today's Activity

Open the notebook for today's activity:

<https://dannycaballero.info/phy321msu/resources/vscode-setup-numeric-activity.html>

Or find it in the course website under **Resources**.

Download the `.ipynb` file and open it in VS Code.

Libraries We'll Use

```
import numpy as np          # Numerical computations
import matplotlib.pyplot as plt # Plotting
import pandas as pd         # Data manipulation
```

Run the first code cell to load these libraries.

If you get an error, you may need to install them:

```
pip install numpy matplotlib pandas
```

The Physics: Numerical Derivatives

In experiments, we measure **positions** at different times.

To get velocities or accelerations, we **numerically differentiate** our data.

$$v_i \approx \frac{x_{i+1} - x_i}{\Delta t}$$

This is the **forward difference** method.

Finite Difference Methods

Three common methods:

- **Forward:** $f'(x_i) \approx \frac{f(x_{i+1}) - f(x_i)}{\Delta x}$
- **Backward:** $f'(x_i) \approx \frac{f(x_i) - f(x_{i-1})}{\Delta x}$
- **Central:** $f'(x_i) \approx \frac{f(x_{i+1}) - f(x_{i-1})}{2\Delta x}$

Central difference is more accurate but uses both neighboring points.

The Synthetic Data

We've generated position data for a ball tossed with air resistance:

$$\mathbf{F}_{\text{net}} = -m\mathbf{g} - C_D|\mathbf{v}|\mathbf{v}$$

Run the data generation cells in the notebook to create:

- `t` - time array (seconds)
- `x` - x-position array (meters)
- `y` - y-position array (meters)

Your Task

Using only finite difference methods:

1. Compute and plot the **numerically-derived velocities** (v_x, v_y)
2. Compute and plot the **numerically-derived accelerations** (a_x, a_y)
3. Compare your numerical derivatives with the "true" values
4. What do you notice about the accuracy?

Work through the notebook with your neighbors!

Hints

To compute velocity from position:

```
dt = t[1] - t[0] # time step  
vx = (x[1:] - x[:-1]) / dt # forward difference
```

Note: Your derivative arrays will be **one element shorter** than the original!

To plot against the right time values:

```
plt.plot(t[:-1], vx) # use t[:-1] not t
```

Exporting Notebooks to PDF

For homework submissions on Gradescope:

Option A: VS Code Extension

- Install "Jupyter PDF Export" extension
- Right-click notebook → Export as PDF

Option B: Command Line

```
jupyter nbconvert --to pdf your_notebook.ipynb
```

Option C: Print to PDF

- Use `Ctrl+P` / `Cmd+P` → Print

Reminders

- **Homework 1** is due next Friday (late after Sunday)
- **Help sessions** start next week
 - Fill out the [help session poll](#)
- Complete the [student information survey](#)

Get Started!

1. Download the notebook from the course website
2. Open it in VS Code
3. Work through the cells
4. Complete the analysis task with your neighbors

Ask questions as you go!